

Digital Forensics with FTK Imager

Principles, Workflow, and Practical Reference

This reference explains the core acquisition and analysis workflow of FTK Imager, why each step preserves evidentiary integrity, and how the pieces fit into a standard digital-forensic process.

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Table of Contents

1. Introduction to Digital Evidence Imaging
 2. Creating a Forensic Disk Image
 3. Capturing Volatile Memory (RAM)
 4. Analyzing an Acquired Image
 5. Mounting an Image as a Drive
 6. Custom Content Image with AD Encryption
 7. Decrypting an AD1 Image
 8. Obtaining Protected System Files
 9. Detecting EFS Encryption
 10. Exporting Files
 11. Best-Practice Checklist
- Quick Reference Table

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1. Introduction to Digital Evidence Imaging

Digital forensics relies on one non-negotiable principle: the original evidence must never be altered. A **forensic image** is an exact, sector-by-sector duplicate of a storage device or a snapshot of live memory, captured in a way that can be mathematically proven identical to the source. FTK Imager, developed by AccessData, is one of the most widely used free tools for producing and verifying these images.

The tool matters because courts, auditors, and internal investigators all depend on a documented, repeatable chain of custody. If evidence can be shown to have changed even by a single bit, its value in an investigation or legal proceeding can be challenged. FTK Imager addresses this by generating cryptographic hashes (MD5/SHA1) at the point of acquisition and again at verification, so any discrepancy is immediately detectable.



Fig 1. Evidence handling pipeline — the original is never touched after acquisition.

2. Creating a Forensic Disk Image

Definition: Disk imaging is the process of producing a bit-for-bit copy of a physical drive (the full hardware device) or a logical drive (a single partition), capturing every sector rather than just visible files.

Why It Matters: A normal file copy only grabs what the operating system currently shows you. A bit-for-bit image also captures deleted files, file slack, and unallocated space — the areas where evidence of tampering, deletion, or hidden activity most often survives. Choosing the right format (Raw/DD for pure compatibility, E01 for built-in metadata and compression) affects how the image will be handled by other tools downstream.

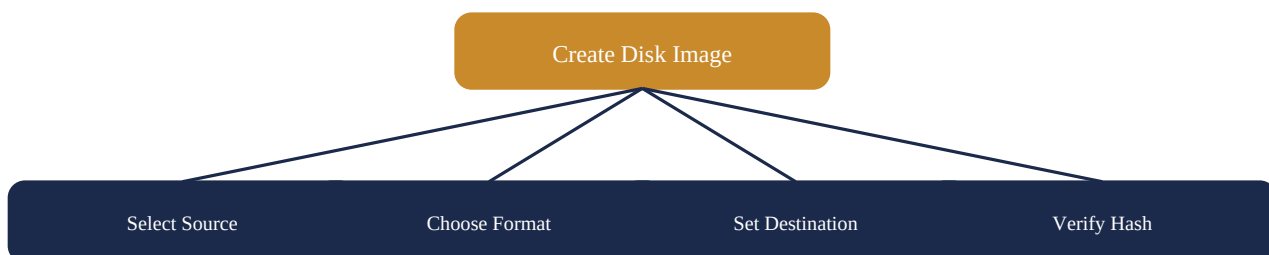


Fig 2. Disk imaging decision tree

Steps:

1. File > Create Disk Image
2. Select source type: Physical Drive or Logical Drive
3. Choose the exact source drive
4. Pick image format: Raw/DD, E01, SMART, or AFF
5. Enter case number, examiner, and notes
6. Set destination and filename, then Start
7. Review the MD5/SHA1 hash verification report

3. Capturing Volatile Memory (RAM)

Definition: Memory capture is the process of dumping the live, running contents of RAM to a non-volatile file, preserving a snapshot of the system's active state at the moment of acquisition.

Why It Matters: RAM is volatile — it holds plaintext passwords, active encryption keys, running malware, open network connections, and unsaved documents, all of which disappear the instant the machine is powered off. Because of this, memory should almost always be captured before a disk image is taken or before a system is shut down; sequencing matters as much as the capture itself.



Fig 3. Memory acquisition flow

Steps:

1. File > Capture Memory
2. Choose destination path and filename
3. Optionally include the pagefile or save as AD1
4. Click Capture Memory and wait for completion

4. Analyzing an Acquired Image

Definition: Analysis is the process of loading a previously created image back into FTK Imager to browse its file system, review the evidence tree, and inspect content in a read-only environment.

Why It Matters: Analysis has to be non-destructive and repeatable, since a second examiner — or opposing counsel — must be able to reach the exact same file structure independently. Viewing the image rather than the original also means any mistake made during review has zero impact on the source evidence.

Steps:

1. File > Add Evidence Item > Image File
2. Browse to the image file
3. Explore the Evidence Tree (includes deleted/unallocated space)
4. Remove the evidence item before loading a new one

5. Mounting an Image as a Drive

Definition: Mounting attaches an image file to the operating system as a virtual, read-only drive letter, making it browsable exactly like a physical disk.

Why It Matters: This lets an examiner point ordinary file browsers or third-party analysis tools directly at the evidence without exporting anything first, while the read-only mount guarantees no write-contamination can occur — a safeguard that write-blockers provide on physical hardware.

Steps:

1. File > Image Mounting
2. Browse to the image, set mount type and method
3. Click Mount — the image appears as a new drive letter

6. Custom Content Image with AD Encryption

Definition: This feature lets an examiner select specific files or folders into a smaller, targeted AD1 image and encrypt that package with a password or a digital certificate.

Why It Matters: Full-disk images are often too large and too sensitive to move around freely. A custom, encrypted AD1 lets an investigator share only the relevant evidence with another party while keeping personal or classified data protected in transit and in storage — directly supporting data-protection and chain-of-custody requirements.

Steps:

1. Right-click chosen files > Add to Custom Content Image (AD1)
2. Click Create Image
3. Enter evidence details and destination
4. Check Use AD Encryption and set a password/certificate

7. Decrypting an AD1 Image

Definition: Decryption reverses AD encryption on a custom content image, using the original password or certificate to produce a readable copy.

Why It Matters: Restricting who can decrypt evidence keeps custody controlled end-to-end: only the credentialed examiner (or someone the case authorizes) can ever view the protected contents, which matters both for legal defensibility and for basic confidentiality of sensitive personal data.

Steps:

1. File > Decrypt AD1 Image
2. Enter the original password
3. Load the resulting decrypted file via Add Evidence Item

8. Obtaining Protected System Files

Definition: This function extracts Windows files that are normally locked while the OS is running, most importantly the registry hives (SAM, SYSTEM, SECURITY, SOFTWARE) and user profile data.

Why It Matters: These hives store password hashes, installed software history, user activity, and system configuration — some of the richest sources of evidence in a Windows investigation — but they are inaccessible through a normal file copy while Windows is running. FTK Imager reaches them directly from the live system, avoiding the need to shut the machine down first.

Steps:

1. File > Obtain Protected Files
2. Browse to a destination folder
3. Select 'Password recovery and all registry files'
4. Click OK

9. Detecting EFS Encryption

Definition: This scan checks a mounted or loaded volume for files protected by Windows' built-in Encrypting File System (EFS) and flags every match.

Why It Matters: EFS-protected files look like ordinary files in a directory listing but are unreadable without the correct certificate or key. Detecting them early prevents wasted analysis effort and tells the investigator exactly which files require a separate decryption step before review can continue.

Steps:

1. File > Detect EFS Encryption
2. Review the flagged results

10. Exporting Files

Definition: Export copies selected files or folders out of a loaded image into a normal, working folder on disk, outside the forensic tool.

Why It Matters: This is the bridge between acquisition and everything that happens next — writing a report, feeding files into other analysis software, or archiving specific artifacts for long-term retention. FTK Imager logs exactly how many files and bytes were copied, giving a verifiable record of what left the evidence store.

Steps:

1. Select files/folders in the Evidence Tree
2. File > Export Files
3. Choose a destination and confirm the export summary

11. Best-Practice Checklist

- Always hash-verify an image immediately after creation, and again before analysis begins.
- Capture volatile memory before powering down or imaging the disk, whenever the system is live.
- Work only from mounted or loaded copies — never analyze the original media directly.
- Encrypt any exported evidence package that will leave a controlled environment.
- Record case number, examiner name, and notes for every image created, without exception.
- Keep a written log of every export: what was taken, when, and by whom.

Quick Reference

Task	Menu Path
Create disk image	File > Create Disk Image
Capture RAM	File > Capture Memory
Load an image	File > Add Evidence Item
Mount image as drive	File > Image Mounting
Encrypted custom content	Right-click files > Add to Custom Content Image
Decrypt AD1	File > Decrypt AD1 Image

Recover system files	File > Obtain Protected Files
Detect EFS	File > Detect EFS Encryption
Export files	File > Export Files

Prepared for personal study and educational reference.

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